

This is a list of NASA missions, both crewed and robotic, since the establishment of NASA in 1958. There are over 80 currently active science missions.

X-Plane program

Since 1945, NACA (NASA's predecessor) and, since January 26, 1958, NASA has conducted the X-Plane Program. The program was originally intended to create a family of experimental aircraft not intended for production beyond the limited number of each design built solely for flight research. The first X-Plane, the Bell X-1, was the first rocket-powered airplane to break the sound barrier on October 14, 1947. X-Planes have set numerous milestones since then, both crewed and unpiloted.

Human spaceflight

NASA has successfully launched over 200 crewed flights. Three have ended in failure, causing the death of the entire crew: Apollo 1 (which never launched) in 1967 lost three crew members, STS-51-L (the Challenger disaster) in 1986, and STS-107 (the Columbia disaster) in 2003.

Notes:

Early Obama administration review

In May 2009, the Obama administration announced the launch of an independent review of planned U.S. human space flight activities with the goal of ensuring that the nation is on a vigorous and sustainable path to achieving its boldest aspirations in space. The review was conducted by a panel of experts led by Norman Augustine, the former CEO of Lockheed Martin, who served on the President's Council of Advisers on Science and Technology under both Democrat and Republican presidents.

The "Review of United States Human Space Flight Plans" was to examine ongoing and planned National Aeronautics and Space Administration (NASA) development activities, as well as potential alternatives and present options for advancing a safe, innovative, affordable, and sustainable human space flight program in the years following Space Shuttle retirement. The panel worked closely with NASA and sought input from the United States Congress, the White House, the public, industry, and international partners as it developed its options. It presented its results on October 22, 2009.

In February 2010, Obama announced his proposal to cancel the Constellation program as part of his reform program. Constellation was officially canceled by the NASA Budget Authorization Act on October 11, 2010.

Artemis program

NASA brought the Orion spacecraft back to life from the defunct Constellation program and successfully test-launched the first capsule on December 5, 2014, aboard EFT-1. After a near-perfect flight traveling 3,600 miles (5,800 km) above Earth, the spacecraft was recovered for study. NASA plans to use the Orion crew vehicle to send humans to deep space locations such as the Moon and Mars starting in the 2020s. Orion will be powered by NASA's new heavy-lift vehicle, the Space Launch System (SLS), which is currently under development.

Artemis I was the first flight of the SLS and was launched as a test of the completed Orion and SLS system. During the mission, an uncrewed Orion capsule spent 10 days in a distant retrograde 60,000 kilometers (37,000 mi) orbit around the Moon before returning to Earth. Artemis II, the first crewed mission of the program, launched four astronauts in April 2026 on a free-return flyby of the Moon at a distance of 8,900 kilometers (5,500 mi).

After Artemis II, the Power and Propulsion Element of the Lunar Gateway and three components of an expendable lunar lander are planned to be delivered on multiple launches

from commercial launch service providers.

Artemis III is planned to launch in 2027 aboard an SLS Block 1 rocket and will use the minimalist Gateway and expendable lander to achieve the first crewed lunar landing of the program. The flight is planned to touch down on the lunar south pole region, with two astronauts staying there for about one week.

Robotic missions

Suborbital

Anomalous Transport Rocket Experiment (ATREX) – five consecutive launches, 80 seconds apart on March 27, 2012, studied the high-altitude jet stream.

NASA Sounding Rocket Program

SHIELDS – launched April 19, 2021, collected data from the heliopause.

Earth and Heliocentric satellites

Applications Technology Satellites

ATS-1, launched December 1966, completed

ATS-2, launched April 1967, completed

ATS-3, launched November 1967, completed

ATS-4, launched August 1968, completed

ATS-5, launched August 1969, completed

ATS-6, launched May 1974, completed

Explorers Program

Explorer 1, launched January 1958, completed

Explorer 2, launched March 1958, failed

Explorer 3, launched March 1958, completed

Explorer 4, launched July 1958, completed

Explorer 5, launched August 1958, failed

Explorer S-1 launched July 1959, failed

Explorer 6, launched August 1959, completed

Explorer 7, Launched October 1960, completed

Biosatellite program

Biosatellite 1, launched December 1966, completed

Biosatellite 2, launched September 1967, completed

Biosatellite 3, launched June 1969, completed

Cosmic Background Explorer (COBE), launched November 1989, completed

Upper Atmosphere Research Satellite (UARS), launched September 1991, completed

Gravity Recovery and Climate Experiment (GRACE), launched March 2002, completed

Gravity Recover and Climate Experiment - Follow-On (GRACE-FO), launched May 2018, operational

NPOESS Preparatory Project (NPP) – National Polar-orbiting Operational Environmental Satellite System (NPOESS), launched October 2011, operational

Echo 1 and 2, launched August 1960 and January 1964, respectively, completed

Great Observatories

Hubble Space Telescope, launched April 1990, operational

Compton Gamma Ray Observatory, launched April 1991, completed

Chandra X-ray Observatory, launched July 1999, operational

Spitzer Space Telescope, launched August 2003, completed

High Energy Astronomy Observatory program

High Energy Astronomy Observatory 1 (HEAO 1), launched August 1977, completed

Einstein Observatory (HEAO 2) launched November 1978, completed – first fully imaging X-ray telescope

High Energy Astronomy Observatory 3 (HEAO 3), launched September 1979, completed

Imager for Magnetopause-to-Aurora Global Exploration (IMAGE), launched March 2000, completed

Infrared Astronomical Satellite (IRAS), launched January 1983, completed

Jason-1, launched December 2001, completed

OSTM/Jason-2, launched June 2008, completed

Jason-3, launched January 2016, operational

Neil Gehrels Swift Observatory (SWIFT), launched November 2004, operational

Landsat program

Landsat 1, launched July 1972, completed

Landsat 2, launched January 1975, completed

Landsat 3, launched March 1978, completed

Landsat 4, launched July 1982, completed

Landsat 5, launched March 1984, completed

Landsat 6, launched October 1993, failed

Landsat 7, launched April 1999, operational

Landsat 8, launched February 2013, operational

Landsat 9, launched September 2021, operational

Living With a Star

Van Allen Probes, launched August 2012, completed – Twin probes studying the Van Allen radiation belt

Earth Observing System

Terra, launched December 1999, operational

Aqua, launched May 2002, operational

Aura, launched July 2004, operational

New Millennium Program (NMP)

Earth Observing-1 (EO-1), launched November 2000, completed

Space Technology 5 (ST5), launched March 2006, completed

Space Technology 6 (ST6)

NanoSail-D, launched August 2008, failed

NanoSail-D2, launched November 2010, completed

Orbiting Carbon Observatory (OCO), launched February 2009, failed

Orbiting Carbon Observatory (OCO-2), launched July 2014, operational

Origins program

Far Ultraviolet Spectroscopic Explorer (FUSE), launched June 1999, completed

Kepler, launched March 2009, completed – searching for Earth-sized exoplanets in the habitable zone

James Webb Space Telescope, launched December 2021, operational

Time History of Events and Macroscale Interactions during Substorms (THEMIS), launched February 2007, operational

Small Explorer program (SMEX)

Aeronomy of Ice in the Mesosphere (AIM), launched April 2007, completed

Fast Auroral Snapshot Explorer (FAST), launched August 1996, completed

Galaxy Evolution Explorer (GALEX), launched April 2003, completed

Interstellar Boundary Explorer (IBEX), launched October 2008, operational

Nuclear Spectroscopic Telescope Array (NuSTAR), launched June 2012, operational – X-ray telescope orbiting Earth

Reuven Ramaty High Energy Solar Spectroscopic Imager (RHESSI), launched February 2002, completed – Sun observing, Earth satellite

Solar Anomalous and Magnetospheric Particle Explorer (SAMPEX), launched July 1992, completed

Submillimeter Wave Astronomy Satellite (SWAS), launched December 1998, completed

Transition Region and Coronal Explorer (TRACE), launched August 1998, completed – Sun

observing, Earth satellite

Wide Field Infrared Explorer (WIRE), launched March 1999, failed

Solar Terrestrial Probes program

Hinode (Solar-B)

Thermosphere Ionosphere Mesosphere Energetics and Dynamics (TIMED)

Two Wide-angle Imaging Neutral-atom Spectrometers (TWINS)

Uhuru

Wilkinson Microwave Anisotropy Probe (WMAP)

Imaging X-ray Polarimetry Explorer (IXPE)

BioSentinel

Carruthers Geocorona Observatory

Nancy Grace Roman Space Telescope, launching May 2027, future

INCUS, launching 2027, future

Sun

Pioneer program

Pioneer 6, 7, 8, and 9, launched between December 1965 and November 1968, completed –

Solar wind, solar magnetic field and cosmic rays

Helios 1 and 2, launched December 1974 and January 1976, completed

Ulysses spacecraft, launched October 1990, completed – ESA partnership

Genesis, launched August 2001, completed – returned sample of solar wind

Solar and Heliospheric Observatory (SOHO), launched December 1995, operational – ESA partnership

Advanced Composition Explorer (ACE), launched August 1997, operational

Solar Maximum Mission (SolarMax), launched February 1980, completed – suffered partial failure after launch; repaired in April 1984 during a Space Shuttle mission

Solar Terrestrial Probes program

TIMED, launched December 2001, operational

Hinode, launched September 2006, operational

STEREO, launched October 2006, operational

MMS, launched March 2015, operational

IMAP, launched September 2025, operational

Parker Solar Probe, launched August 2018, operational – the first mission into the Sun's corona

Living With a Star

Solar Dynamics Observatory (SDO), launched February 2010, operational

Balloon Array for RBSP Relativistic Electron Losses (BARREL) – two campaigns of 20 balloons each, studying the Van Allen radiation belts, 2012 to 2014 This mission is a complement to the Van Allen Probes (RBSP).

CubeSat for Solar Particles (CuSP), launched November 2022

Moon

Pioneer program

Pioneer 0, launched August 1958, failed

Pioneer 1, launched October 1958, failed

Pioneer 2, launched November 1958, failed

Pioneer 3, launched December 1958, failed

Pioneer 4, launched March 1959, completed

Pioneer P-1, launched September 1959, failed

Pioneer P-3, launched November 1959, failed

Pioneer P-30, launched September 1960, failed

Pioneer P-31, launched December 1960, failed

Ranger program

Ranger 1, launched August 1961, failed

Ranger 2, launched November 1961, failed

Ranger 3, launched January 1962, failed

Ranger 4, launched April 1962, failed

Ranger 5, launched October 1962, failed

Ranger 6, launched January 1964, failed

Ranger 7, launched July 1964, completed

Ranger 8, launched February 1965, completed

Ranger 9, launched March 1965, completed

Surveyor program

Surveyor 1, launched May 1966, completed

Surveyor 2, launched September 1966, failed

Surveyor 3, launched April 1967, completed

Surveyor 4, launched July 1967, failed

Surveyor 5, launched September 1967, completed

Surveyor 6, launched November 1967, completed

Surveyor 7, launched January 1968, completed

Lunar Orbiter program

Lunar Orbiter 1, launched August 1966, completed

Lunar Orbiter 2, launched November 1966, completed

Lunar Orbiter 3, launched February 1967, completed

Lunar Orbiter 4, launched May 1967, completed

Lunar Orbiter 5, launched August 1967, completed

Clementine, launched January 1994, completed

Discovery Program

Discovery 3 – Lunar Prospector, launched January 1998, completed

Discovery 11 – GRAIL, launched September 2011, completed

Moon Mineralogy Mapper (M3) – instrument for ISRO's Chandrayaan-1, launched October 2008, completed

Lunar Precursor Robotic Program (LPRP)

LCROSS, launched June 2009, completed

Lunar Reconnaissance Orbiter (LRO), launched June 2009, operational

LADEE, launched September 2013, completed

Korea Pathfinder Lunar Orbiter's ShadowCam instrument, launched August 2022, operational

CAPSTONE, launched November 2022, operational

LunaH-Map, launched November 2022, operational

Lunar IceCube, launched November 2022, operational

Lunar Flashlight, launched December 2022, failed

Commercial Lunar Payload Services

Peregrine Mission One, launched January 2024, failed

IM-1, launched February 2024, completed – first commercial lunar landing. First American Moon landing since Apollo 17 in December 1972.

Blue Ghost Mission 1, launched January 2025, completed

IM-2, launched February 2025, completed

Artemis Program

Artemis I, launched November 2022, completed

Artemis II, launched April 2026, completed

Mercury

Mariner program

Mariner 10, launched November 1973, completed – flyby of Venus; multiple flybys of Mercury;

first spacecraft to Mercury

Discovery program

Discovery 7 – MESSENGER, launched August 2004, completed – first to orbit Mercury

Venus

Mariner program

Mariner 1, launched July 1962, failed – intended to be first American flyby of Venus

Mariner 2, launched August 1962, completed – first flyby of Venus by an operational spacecraft

Mariner 5, launched June 1967, completed – flyby of Venus

Mariner 10, launched November 1973, completed – flyby of Venus; multiple flybys of Mercury; first spacecraft to Mercury

Pioneer program

Pioneer 5, launched March 1960, completed – interplanetary space between Earth and Venus

Pioneer Venus project

Pioneer Venus Orbiter, launched May 1978, completed – Venus orbiter

Pioneer Venus Multiprobe, launched August 1978, completed – Venus atmospheric probes

Magellan, launched May 1989, completed – radar mapping of Venus

Discovery Program

Discovery 15 – VERITAS, launching 2031, future

Discovery 16 – DAVINCI, launching 2030, future

Mars

Mariner program

Mariner 4, launched November 1964, completed

Mariner 6 and 7, launched February 1969, completed

Mariner 8, launched May 1971, failed

Mariner 9, launched May 1971, completed

Viking program

Viking 1, launched August 1975, completed

Viking 2, launched September 1975, completed

Mars Observer, launched September 1992, failed

Mars Global Surveyor, launched November 1996, completed

Discovery Program

Discovery 2 – Mars Pathfinder / Sojourner rover, launched July 1997, completed

Discovery 12 – InSight, launched May 2018, completed

Mars Polar Lander, launched January 1999, failed

Deep Space 2, launched January 1999, failed – (sub-surface probes)

2001 Mars Odyssey, launched April 2001, operational

Mars Exploration Rovers

Spirit rover, launched June 2003, completed

Opportunity rover, launched June 2003, completed

Mars Reconnaissance Orbiter, launched August 2005, operational – Mars orbiter

Mars Scout program

Phoenix, launched August 2007, completed – Mars lander

MAVEN, launched November 2013, completed – orbiter that studied the atmosphere of Mars

Mars Science Laboratory (MSL) / Curiosity rover, launched November 2011, operational – Mars rover exploring Gale Crater

InSight, launched May 2018, completed

Mars 2020

Perseverance rover, launched July 2020, operational – Mars rover exploring Jezero Crater

Ingenuity helicopter, launched July 2020, completed – first powered flight on Mars

EscaPADE, launched November 2025, enroute to mars.

Jupiter

Pioneer program

Pioneer 10, launched March 1972, completed – first to the asteroid belt and Jupiter

Pioneer 11, launched April 1973, completed – asteroid belt and Jupiter, first to Saturn

Voyager program

Voyager 1, launched September 1977, operational – flybys of Jupiter and Saturn; extended mission to explore interstellar medium; most distant human-made object

Voyager 2, launched August 1977, operational – flybys of Jupiter, Saturn, Uranus, and Neptune; extended mission to explore interstellar medium; first spacecraft to Uranus and Neptune

Galileo, launched October 1989, completed – Jupiter and its moons

New Frontiers program

New Frontiers 2 – Juno, launched August 2011, operational – Jupiter orbiter mission

Europa Clipper, launched October 2024, enroute

Saturn

Pioneer program

Pioneer 11, launched April 1973, completed – asteroid belt and Jupiter, first to Saturn

Voyager program

Voyager 1, launched September 1977, operational – flybys of Jupiter and Saturn; extended mission to explore interstellar medium; most distant human-made object

Voyager 2, launched August 1977, operational – flybys of Jupiter, Saturn, Uranus, and Neptune; extended mission to explore interstellar medium; first spacecraft to Uranus and Neptune

Cassini–Huygens, launched October 1997, completed – Saturn and its moons

New Frontiers program

New Frontiers 4 – Dragonfly, launching 2028, future

Enceladus Orbilander, launching 2038, future

Uranus

Voyager program

Voyager 2, launched August 1977, operational – flybys of Jupiter, Saturn, Uranus, and Neptune; extended mission to explore interstellar medium; first spacecraft to Uranus and Neptune

Uranus Orbiter and Probe, launching 2032, future

Neptune

Voyager program

Voyager 2, launched August 1977, operational – flybys of Jupiter, Saturn, Uranus, and Neptune; extended mission to explore interstellar medium; first spacecraft to Uranus and Neptune

Asteroids/comets

New Millennium Program (NMP)

NEAR Shoemaker, launched February 1996, completed – close study of 433 Eros

Deep Space 1 (DS1), launched October 1998, completed – first spacecraft propelled by an ion thruster

Discovery Program

Discovery 4 – Stardust, launched February 1999, completed – follow-up for Deep Impact's primary mission to 9P/Tempel

Discovery 6 – CONTOUR, launched July 2002, failed

Discovery 8 – Deep Impact (primary); EPOXI (extended), launched January 2005, completed

Discovery 9 – Dawn, launched September 2007, completed – Vesta in 2011-2012, and Ceres in 2015-2018

Discovery 13 – Lucy, launched October 2021, operational – Will fly by one main-belt asteroid and seven Jupiter Trojan asteroids.

Discovery 14 – Psyche, launched October 2023, enroute

New Frontiers program

New Frontiers 3 – OSIRIS-REx – launched September 2016, operational

Double Asteroid Redirection Test (DART), launched November 2021, completed

Near-Earth Asteroid Scout, launched November 2022, failed

Dwarf planets

New Frontiers program

New Frontiers 1 – New Horizons, launched January 2006, operational – flyby of Pluto and its moons in 2015; first to Pluto

Discovery Program

Discovery 9 – Dawn, launched September 2007, completed – Vesta in 2011-2012, and Ceres in 2015-2018

Canceled or undeveloped missions

Comet Rendezvous Asteroid Flyby (CRAF)

Jupiter Icy Moons Orbiter (JIMO)

Mars Astrobiology Explorer-Cacher (MAX-C)

Mars Telecommunications Orbiter (MTO)

Asteroid Redirect Mission (2013–2017)

Origins Program

Space Interferometry Mission (SIM)

Terrestrial Planet Finder (TPF)

Pluto Kuiper Express (PLUTOKE) – replaced by New Horizons

Triton Hopper (Rocket-powered Hopper)

Old proposals

Mars Scout program

Aerial Regional-scale Environmental Survey (ARES) (2000-2010 concept)

TAU (spacecraft)- probe to 1000 AU (1980s concept)

See also

NASA:

Large strategic science missions, the NASA flagship missions

Discovery Program, medium-cost NASA missions

New Frontiers program, medium-large NASA missions to outer planets

When We Left Earth: The NASA Missions – 2008 documentary covering NASA's mission history.

Space exploration

Timeline of Solar System exploration

List of European Space Agency programmes and missions

List of ISRO missions

References

Bibliography

Grush, Loren (May 17, 2019). "NASA administrator on new Moon plan: 'We're doing this in a way that's never been done before'". The Verge. Retrieved January 13, 2023.

Heaton, Andrew; Sood, Dr. Rohan (August 10, 2020). "Space Launch System Departure Trajectory Analysis for Cislunar and Deep-Space Exploration" (PDF). NASA. p. 7. Retrieved January 28, 2023.

External links

[NASA - Missions](#)